

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of ECE/CSE/IT/PHY/MTC

End-Term Examination May – 2026

Course: B. Tech

Semester:- 2nd

Branch: All

Subject Code:- CSE/CSE(CPS)/CSE(CS)/ CSE(DS)/CSE(AI)/ECE/IT/ PHY/MTC-2003

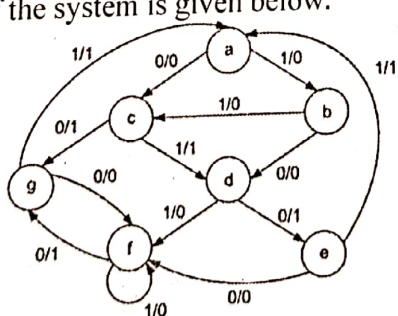
Max. Marks:60

Subject Name: Digital Logic Design (DLD)

Time: 03 Hours

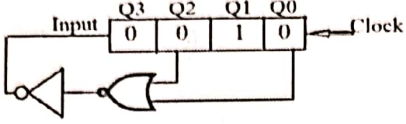
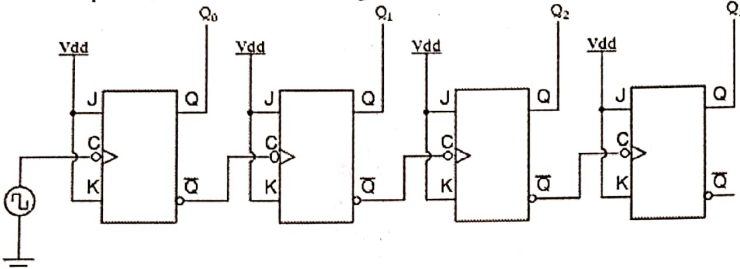
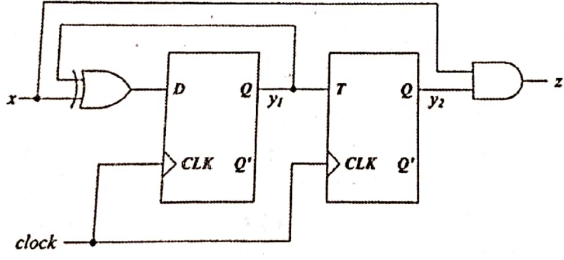
NOTE:

1. All the questions are compulsory.
2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.
4. Calculator is not allowed.

1.A	Two friends are playing ludo with one dice, when they started playing initial three dice numbers after rolling are “four”, “six”, “five”. They moved the pieces as double of the numbers and the consecutive three rolls the numbers are ‘three, one and two’ they move the pieces on the board as the square of the numbers. (i) Obtain the truth table by finding out the number of input and output variables for the above scenario. (ii) Find a simplified Boolean expression using a K-Map.	2+3
B.	In a busy urban intersection, a traffic light control system is designed using multiplexers to manage the flow of vehicles and pedestrians efficiently. The system employs multiplexers to select and control the timing of traffic light signals based on real-time traffic conditions, pedestrian crossings, and emergency vehicle prioritization. How are multiplexers configured to route traffic light signals between different phases (e.g., Green, Yellow, Red) at an intersection?	5
2.A	In a multi-level parking area, a digital display shows the number of available parking slots. Due to maintenance constraints, only 5 slots are operational, and the display cycles through the available slot numbers in descending order: $7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 7 \rightarrow \dots$ Design a synchronous sequential circuit using JK flip-flops to implement this system efficiently.	5
B.	A smart elevator system in a building is designed using a finite state machine (FSM). Different states represent various operations such as idle, moving, door opening/closing, and emergency conditions. The transitions between states depend on input signals like user requests and system conditions. The state diagram of the system is given below.  (a) Write the state table corresponding to the given state diagram. (b) Identify and eliminate equivalent states and obtain the reduced state table. (c) Draw the reduced state diagram.	2+2+1
3.A	(i) Realize AND, OR logic gate using CMOS as a switching circuit. (ii) Explain the following terms: Fanout, Propagation Delay, Noise Margin.	2+3

(Please Turn Over)



B.	In a digital communication system, a 4-bit shift register is used to transmit binary data serially from one device to another. The register performs the right shift operation on every clock pulse, as shown in the figure.  Initially, the data was stored in the register $Q_3Q_2Q_1Q_0 = 0010$. Write the contents of the shift register $Q_3Q_2Q_1Q_0$ after 6 clock pulses. Explain output after each clock pulse.	5															
4.A.	In decorative lighting systems, a chasing LED pattern is created using sequential circuits. The given circuit represents a chain of flip-flops used to generate a moving light pattern, in which the output shifts with each clock pulse, as shown in the Figure (assume initial states as 0000).  (a) Draw the timing diagram showing the outputs Q_0, Q_1, Q_2, Q_3 for at least 5 clock pulses. (b) Identify the type of sequential circuit implemented (e.g., register, counter, etc.) and justify your answer.	3+2															
B.	Convert a negative-edge-triggered T flip-flop to XY flip-flop defined by the characteristic table shown below. <table border="1" data-bbox="574 873 1005 1041"> <thead> <tr> <th>X</th><th>Y</th><th>Q_{n+1}</th></tr> </thead> <tbody> <tr> <td>Gnd</td><td>Gnd</td><td>Q'</td></tr> <tr> <td>Gnd</td><td>+Vcc</td><td>1</td></tr> <tr> <td>+Vcc</td><td>Gnd</td><td>0</td></tr> <tr> <td>+Vcc</td><td>+Vcc</td><td>Q</td></tr> </tbody> </table>	X	Y	Q_{n+1}	Gnd	Gnd	Q'	Gnd	+Vcc	1	+Vcc	Gnd	0	+Vcc	+Vcc	Q	5
X	Y	Q_{n+1}															
Gnd	Gnd	Q'															
Gnd	+Vcc	1															
+Vcc	Gnd	0															
+Vcc	+Vcc	Q															
5.A	Analyse the synchronous sequential circuit and derive its state table and state diagram. 	5															
B.	In a smart building automation system, a central controller needs to send control signals to one out of 16 different devices (such as lights, alarms, sensors, or appliances). To efficiently route the signal, a demultiplexer-based design is used. However, due to hardware constraints, only 2:4 decoders with enable inputs are available for implementation. Design a 1:16 demultiplexer using only 2:4 decoders with enable inputs. (No additional logic gates are allowed.)	5															
6.A	A new Binary Coded Pentary (BCP) number system is proposed in which every digit of a base-5 number is represented by its corresponding 3-bit binary code. For example, the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 10001001011 corresponds of the which number is base-5 system.	5															
B.	(i) A company has $(13)_r$ departments, and each department has $(24)_r$ number of employees. Company is also having $(41)_r$ security guards. The total strength of the company including security guards is $(1013)_r$. What is the value of r ? (ii) Perform following operations using 2's complement method (represent in 8-bits) (a) $48 - 23$ (b) $23 - 48$	2+3															

*****ALL THE BEST*****